

LEARNOVA

Progress & Research Report

Cognitive-Load-Aware Generation of Educational Slide Decks

A calibrated slide-fitness metric (PSF) and an optimal segmentation algorithm (CLASS)

Prepared for: project owner

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Repository: Kawal511/Learnova (local clone)

Contents

Executive summary

This report documents work on the *Learnova* AI presentation engine over one working session, and sets out a research programme intended to take the project to a top-tier publication. Three things were achieved: (1) the repository was cloned, built, and verified, and a genuine dependency bug was fixed; (2) the project's ad-hoc “engagement score” was replaced with a theory-grounded, implementable research contribution — the **Pedagogical Slide Fitness (PSF)** metric and the **CLASS** slide-segmentation algorithm — with full source, a test suite, and a paper draft; (3) the draft was critically reviewed and a substantially more ambitious research agenda was designed.

Headline research idea: a slide generator should be judged and built by the *same* model. If we can score a slide's cognitive load, we can also paginate content to minimise it. PSF is that score; CLASS is the optimiser that uses it. CLASS is a generalisation of Knuth & Plass optimal line breaking, is provably optimal, and can never drop source content.

Status: PSF and CLASS are implemented, tested (16 new unit tests, incl. a brute-force optimality check), and wired into the pipeline behind opt-in switches so nothing existing is disturbed — 305 of the project's existing tests still pass. The empirical study (human ratings, learning outcomes) is designed but not yet run.

Part I · Work completed

1. Repository and environment

- Cloned Kawal511/Learnova exactly; git history intact (HEAD = 78087d3).
- Created an isolated virtual environment and installed the full dependency set (core pipeline + Streamlit UI + FastAPI backend + test tooling).
- Note for reproduction: the dependency build must run outside the sandboxed shell — building PyMuPDF is memory-heavy and the sandbox OOM-kills it (exit 137).

2. Bug found and fixed

Missing dependency. The Clerk authentication module (`src/learnova/auth/clerk.py`) imports `jwt` (PyJWT), but PyJWT appears in neither `requirements.txt` nor `pyproject.toml`. The test suite could not even be collected (`ImportError` in `test_theme_and_storage.py`). Added `PyJWT>=2.8.0` to both files.

3. Baseline verification

Check	Result
Offline test suite (pytest -m “not live”)	305 passed, 4 skipped — green
Pipeline, typed input, no API keys	Produces .pptx + interactive HTML via heuristic fallback
End-to-end smoke (photosynthesis syllabus)	3 slides, engagement 84/100, both artifacts built
Post-change re-run (PSF + CLASS added)	305 + 16 new = 321 passed; CLASS path also green

Part II · Research contribution (implemented)

4. Why the existing scorer is not publishable

The current engagement scorer (`scoring/scorer.py`) assigns a 0–100 score as a fixed weighted sum of six hand-chosen sub-scores (text density 20, bullet count 20, title quality 15, takeaway 15, readability 15, visual bonus 15). The weights and the piecewise thresholds are asserted — not derived from any theory, not fitted to data, and never validated against a human judgement of learning quality. It is a reasonable engineering heuristic and a weak research contribution. Worse, its additive form lets a beautiful empty slide score ~66/100.

5. The PSF metric

PSF scores a slide s in $[0, 1]$ as a **multiplicative** model of three sub-indices, each derived from an established learning theory:

$$PSF(s) = E(s)^{\alpha} \cdot (1 - L(s))^{\beta} \cdot C(s)^{\gamma}, \quad \alpha + \beta + \gamma = 1$$

Sub-index	Meaning	Theoretical basis
$E(s)$ information efficiency	Novel content delivered, with diminishing returns and a redundancy discount	Shannon information / idf-weighted content
$L(s)$ cognitive load	Logistic function of load drivers: element interactivity, extraneous text, visual complexity, split attention	Cognitive Load Theory (Sweller)
$C(s)$ multimedia coherence	Mean of four measurable indicators: signalling, coherence, non-redundancy, spatial contiguity	Mayer's Cognitive Theory of Multimedia Learning

5.1 Information efficiency

```
U(s) =  $\sum \text{idf}(c)$  over content concepts  $c$  first introduced on slide  $s$   
R(s) = mean bullet overlap with the title and the other bullets (Jaccard)  
E(s) = ( U(s) / (U(s) +  $\kappa_E$ ) ) · (1 - R(s))
```

The saturating term $U/(U+\kappa_E)$ encodes “a slide has one job”: the first few rare concepts earn most of the score. A slide with a title but no body or figure delivers nothing, so $E = 0$ — which is exactly the empty-slide failure the old scorer rewards.

5.2 Cognitive load

```
 $\hat{e}$  =  $n_{\text{concepts}} \cdot (1 + \text{relational\_cue\_density}) / e_{\text{ref}}$  (intrinsic / interactivity)  
 $\hat{t}$  =  $\text{words} / t_{\text{ref}}$  (extraneous text load)  
 $\hat{v}$  =  $\max(\text{table\_rows}, \text{flow\_nodes}, \text{grid\_cards}) / v_{\text{ref}}$  (visual complexity)  
 $\hat{s} = 1$  if a figure is described in text but placed on another slide, else 0  
 $L(s) = \sigma( w_e \cdot \hat{e} + w_t \cdot \hat{t} + w_v \cdot \hat{v} + w_s \cdot \hat{s} - \theta_L )$ 
```

Relational-cue density (counts of “because”, “therefore”, “whereas”, “leads to”...) is a text proxy for how many elements interact — which CLT says drives intrinsic load, not raw item count.

5.3 Deck aggregation

$$PSF_{\text{deck}} = (\prod_s PSF(s))^{(1/N)} \cdot \text{Flow}(\text{deck})$$

Flow is a trapezoidal reward for moderate concept overlap between consecutive slides: near 0 is a jarring topic jump, near 1 is repetition.

5.4 Calibration — the key methodological point

Only a low-dimensional weight vector ($\alpha, \beta, \gamma, w_e, w_t, w_v, w_s, \theta_L$) is learned, by constrained regression against mean human ratings under cross-validation. The scale constants ($\kappa_E, e_{\text{ref}}, t_{\text{ref}}, v_{\text{ref}}$) are fixed from theory (working-memory capacity) and corpus statistics, which keeps the model identifiable. The shape is theory; only the weights are data.

6. The CLASS algorithm

CLASS (Cognitive-Load-Aware Slide Segmentation) decides how a section's ordered content blocks are cut into slides. Today the pipeline uses a fixed per-profile cap plus an even rebalance — a greedy choice that ignores that slide difficulty is non-linear in length.

6.1 Formulation

$$\text{minimise } \sum_j \text{cost}(\text{group}_j) + \lambda \cdot k \quad \text{s.t. } \text{group size} \leq P_{\text{max}}$$

$$\text{cost}(g) = L(\text{slide}(g)) + \mu \cdot (1 - E(\text{slide}(g))) + v \cdot \text{overflow}(g)^2$$

This is structurally identical to Knuth & Plass optimal line breaking: “line” → “slide”, “line badness” → “slide cognitive load”, “hyphenation penalty” → λ . The $\mu \cdot (1 - E)$ term stops the optimiser from emitting near-empty slides.

6.2 Algorithm

```
CLASS(blocks b[1..m], P_max, λ):
    best[0] = 0
    for i in 1..m:
        best[i] = +∞
        for p in max(0, i - P_max) .. i-1:
            c = best[p] + cost(b[p+1..i]) + λ
            if c < best[i]: best[i] = c ; cut[i] = p
    follow cut[] back from m to recover the groups

time  O(m · P_max)      space  O(m)      (P_max ≤ 8, so linear in practice)
```

6.3 Guarantees

- **Completeness:** cut points partition 1..m, so every block lands on exactly one slide. Truncation is structurally impossible — unlike a bullets[:4] cap.
- **Optimality:** global minimum of the objective over all legal segmentations (standard DP exchange argument, Knuth–Plass §3).
- **No orphan slides:** because cost includes (1-E), the optimiser will not emit a one-line continuation slide unless P_max forces it — fixing the bug the even-split code patches with an ad-hoc rebalance.

7. Implementation and tests

File	Role
src/learnova/scoring/psf.py	PSF sub-indices, deck aggregation, and the CLASS DP. Pure stdlib, deterministic, no LLM.
src/learnova/scoring/scorer.py	score_all_slides(deck, engine="psf") — opt-in PSF path beside the original heuristic
src/learnova/pipeline/density.py	LEARNOVA_USE_CLASS=1 switches pagination from even-split to load-optimal CLASS
scripts/calibrate_psf.py	Fits PSFParams from a human-ratings CSV; ships literature-derived priors
tests/test_psf.py	16 unit tests incl. brute-force optimality check for CLASS

File	Role
docs/research/PSF_DESIGN.md	Full design: math, algorithm, evaluation plan, novelty statement
docs/research/paper_draft.md	Paper draft: abstract, sections, RQ table to fill after data collection

Part III · Critical review of the current draft

8. What is strong

- The unifying thesis — the metric is also the cost function of the algorithm that optimises the artefact. That is a clean, memorable spine for a paper.
- The Knuth–Plass generalisation is elegant, citable, and comes with a real optimality proof and a checkable content-preservation property.
- Fully deterministic, no LLM, millisecond runtime — trivially reproducible, which reviewers increasingly demand.
- Grounded in three established theories rather than invented from scratch.

9. Weaknesses a reviewer will attack

#	Objection	Fix
1	The empirical core is entirely “TBD” — 100% method, 0% results.	Collect the corpus + human ratings; report cross-validated numbers before submission.
2	The rarity estimate “idf from word length” is embarrassing for a research paper.	Use real corpus idf (Wikipedia / SUBTLEX) as primary; keep the length heuristic only as a robustness ablation.
3	No LLM baseline — the obvious question “can’t GPT-5 just make good slides?” is unanswered.	Add: LLM-generated decks scored by PSF and by humans; and PSF as a re-ranker over LLM outputs (best-of-N).
4	Circularity: CLASS minimises a cost built from L and E, then decks are evaluated with PSF, which contains L and E.	Primary evaluation of CLASS must be independent: human preference, self-paced reading time, learning outcomes.
5	Too many hand-set knobs (Cobb–Douglas exponents, flow band, τ).	Fit what can be fit; run a full sensitivity analysis for the rest.
6	CLT is tripartite (intrinsic / extraneous / germane); the model measures only 2 of 3.	Add a germane-load term G (worked examples, self-explanation prompts, retrieval-practice quizzes).
7	“Element interactivity via connective counting” is an unvalidated proxy.	Validate against human-coded interactivity ratings on a subsample.
8	One language, text-centric; figure-heavy STEM decks untested.	Add a cross-domain stratum and a figure-heavy condition; report per-stratum correlation.
9	Optimising for measured load could yield dumbed-down decks that feel good but teach less.	The learning-outcome study (E5) must be run, not optional — it is the safeguard.
10	Related work is thin.	Add: DOC2PPT / AutoPresent / SlideGen, prior slide-quality rubrics, Knuth–Plass, rate–distortion in sentence compression, Paas / Leppink cognitive-load measurement.

Part IV · A more ambitious research programme

10. The reframing: Working-Memory-Bounded Presentation Generation

To lift this from “a metric and an algorithm” to a framework paper, cast the whole problem as lossy semantic source coding through a cognitively bounded channel:

Coding concept	Presentation-generation counterpart
Source	The document's semantic content — a concept-dependency graph
Encoder	The slide generator
Channel	The learner's working memory — bounded capacity (≈ 4 chunks) per slide, sequential
Distortion	Semantic content lost — constrained to 0 at the concept level (“nothing dropped”)
Rate	Number of slides (or total presentation time)

Under this lens, PSF is the operational objective, CLASS is the rate-distortion-optimal encoder for the segmentation sub-problem, and the headline evaluation artefact becomes a **rate-distortion curve for presentations** — semantic fidelity versus slide count, plotted for CLASS vs greedy vs LLM.

11. Tier 1 additions — do these for the main paper

Addition	What it is	Why it lands
Germane-load term G	Extend PSF to E / L / G, matching CLT exactly. G measures worked examples, self-explanation prompts, and retrieval-practice quizzes — pulling Learnova's existing inline-quiz feature into the model as a measured variable.	Closes the most obvious theory gap; makes an existing product feature a research result.
PSF-guided decoding	Use PSF as a reward to re-rank N LLM-generated slide variants (best-of-N), and optionally to fine-tune a small slide-writer with DPO / GRPO.	Answers “why not just use an LLM” and shows the metric is useful, not merely descriptive.
Learner-adaptive CLASS (CLASS-A)	Make channel parameters (P_max, load weights) depend on the learner's prior knowledge, measured by pretest or knowledge-graph overlap. The optimal segmentation then changes per learner — the expertise-reversal effect, operationalised.	A second algorithmic result; strong personalised-learning angle.
Concept-dependency DAG + ordering	Extract a prerequisite DAG among concepts; measure prerequisite-violation rate; reorder slides by a minimal-disruption topological sort.	A third optimisation problem; connects to knowledge-space theory.
Rate-distortion evaluation	Plot semantic fidelity vs slide count for every method; report the achievable region and where LLMs sit relative to the CLASS frontier.	One figure that carries the paper.

12. Tier 2 additions — stretch / follow-up paper

- **Cognitive-load budget allocation.** A deck has a total attention budget B; allocate it across slides (convex / knapsack) so hard concepts get more “room”. Whole-deck optimisation, not slide-by-slide.

- **Behavioural validation.** Predict per-slide dwell time and backtracking from L in a self-paced web reading study; correlate predicted load with observed reading behaviour (a cheap proxy for eye-tracking).
- **Theoretical results for CLASS.** Beyond optimality: a construction where greedy even-split is arbitrarily worse than optimal, and a stability bound (a small content edit changes the segmentation by a bounded amount).
- **Adversarial robustness.** Show PSF is not gamed by keyword stuffing (saturation + redundancy penalty) or by padding — an explicit adversarial-evaluation section.
- **Multi-objective Pareto front.** PSF vs brevity vs fidelity — expose the front, produce a family of decks, report hypervolume.
- **Cross-lingual / cross-modal transfer.** Does a PSF calibrated on English text-decks transfer to another language, or to figure-heavy decks?

13. Revised evaluation plan

ID	Question	Method	Primary metric
E1	Does PSF predict human-judged slide quality better than the baseline and strong ablations?	5-fold CV, held-out test set	Spearman ρ , RMSE, ranking NDCG
E2	Which terms matter?	Leave-one-term-out ablation of E / L / G / C and each load driver	Δp
E3	Is the multiplicative form justified vs additive?	Fit both, compare on test	Δ RMSE, calibration plots
E4	Does CLASS beat greedy even-split?	Within-subjects: same content, blind human preference + reading time	Preference %, Δ reading time
E5	Do CLASS decks improve learning, not just perception?	Between-subjects quiz-score study	Mean post-test score, time-on-task
E6	Does PSF-guided re-ranking beat raw LLM decks?	Best-of-N vs N=1, human eval	Preference %, Δ PSF, Δ post-test
E7	Does CLASS-A help the learners it targets?	Novice vs expert $\times$ adaptive vs fixed	Interaction effect on post-test
E8	Runtime	CLASS vs even-split across section lengths	Milliseconds
E9	Is PSF robust to gaming?	Adversarial padding / keyword stuffing	Score inflation vs human score

Part V · Deliverables and next steps

14. Files delivered this session

- `src/learnova/scoring/psf.py` — PSF metric + CLASS algorithm (new)
- `tests/test_psf.py` — 16 tests, all passing (new)
- `scripts/calibrate_psf.py` — parameter-fitting harness (new)
- `docs/research/PSF_DESIGN.md` — design document (new)
- `docs/research/paper_draft.md` — paper draft (new)
- `docs/research/README.md` — research-track index (new)
- `src/learnova/scoring/scorer.py` — opt-in engine=“psf” path (modified)
- `src/learnova/pipeline/density.py` — opt-in CLASS pagination (modified)
- `requirements.txt`, `pyproject.toml` — PyJWT dependency fix (modified)

15. Recommended next steps, in order

1. Decide scope: main-paper tier-1 set (Section 11) vs a leaner first paper (PSF + CLASS only).
2. Swap the length-based rarity estimate for real corpus idf; re-run the internal sanity checks.
3. Add the germane-load term G and fold in the inline-quiz feature.
4. Build the deck corpus from open educational documents (OpenStax, MIT OCW, arXiv survey intros); target 150–300 slides, stratified by discipline.
5. Run the human-rating study (≥ 3 raters/slide); report inter-rater reliability.
6. Calibrate PSF; run E1–E3.
7. Run the CLASS study (E4) and, if resourced, the learning study (E5).
8. Add PSF-guided re-ranking over an LLM baseline (E6).
9. Fill the results tables in `paper_draft.md`, write related work, submit.
10. Separately: begin the web-first “render-then-capture” architecture (HTML deck as source of truth → headless render → rebuild PPTX from the rendered layout).